

AIR COMPRESSOR PREDICTIVE MAINTANENCE DATASET

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What is the purpose?

What is the purpose of the dataset and the case study?

Classified datasets are required for predictive maintenance. A machine system has many parts that are difficult to replace and maintain. When these parts are corrupted, the trained neural network should be able to predict with high accuracy which part is corrupted. That's why as much data is collected as possible. Some data may be fully correlated with each other. This data is still taught to the neural network because changing one parameter in the time domain can unexpectedly change other parameters. In the artificial intelligence system required for predictive maintenance, there must be LSTM next to DNN.

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This data set has been prepared with measurements made on the compressor system feeding the air line of a factory. The related compressor has the characteristics of being driven by an AC current electric motor, two-pistons, water-cooled, single-stage, capable of producing maximum 8 bar compressed air.

Measurements were made with high resolution sensors and an industrial type data collector. To prepare a clean dataset, measurement lines with cable-induced noise were deleted.

What is Air Compressor?

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An air compressor is a device that converts power (using an electric motor, diesel or gasoline engine, etc.) into potential energy stored in compressed air. The compressed air can then be used to power pneumatic tools, inflate tires, and operate various other mechanical devices. Air compressors come in a variety of sizes and types, including portable, stationary, and industrial models. They can be powered by electricity, gasoline, or diesel fuel.

What is Predictive Maintenance

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Predictive maintenance is a technique used to predict when equipment or machinery is likely to fail, so that maintenance can be scheduled in advance. The goal of predictive maintenance is to prevent equipment failure and minimize downtime, by proactively identifying and addressing potential issues before they occur. This is typically accomplished by collecting and analyzing data from sensors on the equipment, such as vibration or temperature readings, and using machine learning algorithms to detect patterns or anomalies that indicate an impending failure.

What we have? (Parameters List)

what we have?

Measured Parameters

- MOTOR RPM
- MOTOR POWER
- MOTOR TORQUE
- COMPRESSOR OUTLET PRESSURE
- PRESSURED LINE AIR FLOW
- COMPRESSOR NOISE
- PRESSURED AIR LINE OUTLET TEMPERATURE
- WATER PUMP OUTLET TEMPERATURE
- WATER INLET TEMPERATURE
- WATER OUTLET TEMPERATURE
- WATER PUMP POWER
- WATER FLOW
- OIL PUMP POWER
- OLI TANK TEMPERATURE [sic]
- GROUND ACCELERATION X,Y,Z
- HEAD CAP ACCELERATION X,Y,Z

Result & Observation Parameters

- BEARING CONDITION
- WATER PUMP CONDITION
- RADIATOR CONDITION
- OUTLET VALVE CONDITION
- MOTOR CONDITION

Motor RPM

The air compressors feeding the factory line are usually driven by an electric motor.

The motor can be connected to the compressor in the form of a belt pulley mechanism or a shaft and coupling.

RPM (rev/min) value refers to the number of revolutions per minute of the electric motor.

**RPM, DATASET, B COLUMN*

Motor Power

Electric motor power refers to the amount of electrical energy that a motor can convert into mechanical power, which is used to drive machinery or equipment. It is typically measured in units of watts (W) or kilowatts (kW), and represents the rate at which the motor can perform work. The power of an electric motor is determined by various factors, including the design of the motor, the amount of electrical current supplied to it, and the efficiency of the motor in converting electrical energy into mechanical energy

**MOTOR_POWER, DATASET, C COLUMN*

Motor Torque

The torques of electric motors are different from those of internal combustion engines.

Internal combustion engines must go above a certain speed to produce torque. At the same time, combustion engines are proportional to power and torque.

Electric motors can produce torque at 0 rpm. Torque can be roughly defined as the turning force of a one-metre rod required to hold a 1kg mass constant. Its unit is given as N/m (Newton/Meter).

**TORQUE, DATASET, D COLUMN*

Here is the transcription of the text from the remaining images, organized by slide.

Noise

Air compressor noise refers to the level of sound produced by an air compressor during operation. It can be measured in units such as decibels (dB) and is influenced by various factors such as the design and size of the compressor, the speed of the motor, the type of compressor technology used, and the acoustic environment in which the compressor is operating. Excessive noise from an air compressor can be a nuisance to nearby workers and can even pose a risk to their hearing if the noise levels are high enough.

Generally, if the sound of the air compressor has increased after a certain run time, it is because the bearings have failed.

**NOISE_DB, DATASET, G COLUMN*

Comp Outlet Temperature

Air compressor outlet air temperature refers to the temperature of the compressed air as it exits the compressor. The outlet air temperature is influenced by various factors such as the temperature of the air being compressed, the compression ratio, the efficiency of the compressor, and the type of compressor technology used.

In general, the outlet air temperature of a compressor will be higher than the ambient air temperature due to the compression process, which generates heat. However, excessively high outlet air temperatures can be a sign of issues such as overheating, inadequate cooling, or inadequate lubrication, which can damage the compressor or reduce its efficiency. Monitoring the outlet air temperature can help to detect potential issues and ensure that the compressor is operating within safe and efficient parameters.

**OUTLET_TEMP, DATASET, H COLUMN*

Image 15: Water Pump Outlet Pressure

Water Pump Outlet Pressure

Compressors compress the air and the compressed air heats up. Metal parts are good heat conductors. In order not to damage the gaskets and to increase the air efficiency, the metal parts must be cooled. Fin or liner is preferred for cooling. Fin compressors do not have liquid cooling. It is cooled by the fan placed close to the compressor. In jacket-cooled models, the combustion chamber is cooled by water.

The energy of the heated water is discharged through a radiator. Gear and centrifugal pumps are used to circulate the water.

**WPUMP_OUTLET_PRESSURE, DATASET, I COLUMN*

Ground Acceleration

Acceleration is the rate at which an object's velocity changes over time.

Mathematically, acceleration can be expressed as the change in velocity divided by the change in time. The SI unit for acceleration is meters per second squared (m/s^2).

Ground acceleration is the acceleration value measured from where the compressor is attached to a rigid part or construction. Acceleration data is measured by the acceleration sensor. The magnitude of the acceleration is related to how much displacement the measurement point shows. Steels create great stresses at large displacements. With great stresses, they form fatigue and cracks in a short time. An increase in ground acceleration over time indicates wear or clearance in the crank, bearing, or caps.

**GAACX,GAACY,GAACZ, DATASET, P,Q,R COLUMNS*

Head Acceleration

Head acceleration is the value measured from the compressor head bolt or upper cooling fin.

Each piece has a natural frequency and they oscillate at a certain frequency as a result of the incoming drive. This oscillation is related to the rigidity of the structure.

Slag accumulates in the outlet valves of the compressors whose air filter deteriorates and which absorbs dust, and the rigidity of the structure changes. The sound of the compressor and the head acceleration change with the change in stiffness. The increase or decrease of these values depends on the type of compressor. However, monitoring this value is a good indicator of the stability of the outlet valve.

**HAACX,HAACY,HAACZ, DATASET, S,T,U COLUMNS*

Indicator Classes

- Bearing
- Water Pump

- Radiator
- Exhaust Valve
- AC Motor

Indicator Classes: Exhaust Valve (Reed)

The most malfunctioning parts in air compressors with reed valves are: Reed valve, bearings, air and oil ring, klingrit gasket and bypass spring.

When the reed valve approaches failure, we can understand this with the measured data. however, when the seal is broken, oil, water and air mix and damage your entire system. Therefore, the maintenance of the gasket is not predictive but preventive.

Water Inlet & Outlet Temperature

In air compressors that are cooled by the circulation of water from the jacket, the water inlet temperature occurs according to the radiator size and fan capacity. The water outlet temperature is related to how much of the heat generated by the heating of the compressed air is transferred to the water. The fact that the water inlet temperature is close to the water outlet temperature indicates that the radiator is dirty.

**WATER_INLET_TEMP, DATASET, J COLUMN *WATER_OUTLET_TEMP, DATASET, K COLUMN*

Comp Outlet Pressure

Atmospheric pressure in the world is around 1 bar. If we compress the air in a closed volume, its pressure increases and at the same time the air heats up.

The air compressor takes air with a piston and tries to force it into a tank. As this cycle continues, the air mass inside the constant volume tank increases. Thus, the pressure of the air increases. The outlet pressure is the value measured just next to the piston valve. Unit: Bar, (100 kPascal)

**OUTLET_PRESSURE_BAR, DATASET, E COLUMN*

Water Pump Power Water Flow

If the radiator is dirty or clogged, the water pump will consume more power as the water pressure will increase. Similarly, if the radiator is dirty, the water flow will decrease.

Unlike internal combustion engines, it is desirable that the coolant entering the compressor be at a minimum temperature. Therefore, the cleanliness and efficiency of the water pump and radiator duo are critical parameters in terms of system efficiency.

**WPUMP_POWER, DATASET, L COLUMN *WATER_FLOW, DATASET, M COLUMN*

Air Flow

Air compressor air flow refers to the amount of air that an air compressor can deliver, typically measured in units such as cubic feet per minute (CFM) or liters per minute (L/min). It represents the volume of air that the compressor can supply at a given pressure, and is an important factor in determining the suitability of a compressor for a particular application. The air flow of an air compressor can vary depending on factors such as the size and design of the compressor, the power source, and the operating conditions, such as the temperature and humidity of the air being compressed. The air flow is typically specified by the manufacturer and can be used to select an appropriate compressor for a specific task.

**AIR_FLOW, DATASET, F COLUMN*

Oil Pump Power Oil Tank Temperature

Pressurized by a pump, the oil passes through fine holes to lubricate cranks, connecting rods, pistons, rings and bearings. The oil also serves to cool the piston. Under normal conditions and at normal operating speed, the oil is not expected to heat up too much. However, if the solid friction is too great, if the wear between parts has started, and the bearings have deteriorated, the oil will overheat. As the oil heats up and the number of particles in the oil increases, the oil pump consumes more power. **OILPUMP_POWER, DATASET, N COLUMN *OIL_TANK_TEMP, DATASET, O COLUMN*